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American University of Phnom Penh

Project Proposal

Topic: Khmer-text-to-speech

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ITM 454 002 - Natural Language Processing

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1. **Problem Statement**

Text-to-Speech (TTS) converts written text into spoken audio. However, developing TTS for Khmer is challenging because Khmer has fewer speech and language resources than high-resource languages. Khmer also has challenges in text processing, such as word and token separation (Kaing et al., 2021). Recently, larger Khmer speech datasets have become available. The Khmer ASR Cultural Dataset provides over 56,000 Khmer speech-text pairs, making it possible to develop Khmer speech applications (Digital Divide Data, 2024). Therefore, this project aims to develop a Khmer TTS system that converts Khmer text into spoken audio.

2. **Proposed Approach**

The purpose of the project is to develop a Khmer Text-to-Speech (TTS) system that converts Khmer written text into spoken audio. This project will use a neural network-based TTS model to train on Khmer speech and transcript data. We will select the dataset that has the consistency from the transcript and the clear voice from the speakers. This system will take the Khmer text as the input and generate spoken Khmer audio as the output.

3. **Methodology**

The project follows several main steps:

- **Dataset:** Using a Khmer speech dataset contains both transcripts and audio recordings that are available on Hugging Face.
- **Text Preprocessing:** Clean normalizing the Khmer text includes Unicode normalization, punctuation handling, and removing some unnecessary characters.
- **Audio Preprocessing:** Prepare the speech recordings throughout checking audio quality and convert to the consistency format.
- **Model Training:** This project will use a VITS neural TTS model. This model will train the paired Khmer text and audio speech. It will learn the relationship between Khmer text and the corresponding speech patterns.
- **Speech Generation:** After training, the VITS model will get new Khmer sentences as the input. Then, it will generate spoken Khmer as the output.
- **Model Evaluation:** The generated speech will be evaluated through Mean Opinion Source (MOS) for naturalness and intelligibility, and Character Error Rate (CER) to measure how accurate the generated speech is, compared to the input text.
- **System Demonstration:** In this project, we will create a simple interface where users can easily enter Khmer text and listen to the speech that is generated by our trained model.

4. **Dataset Description**

For this project, the Khmer ASR Cultural Dataset, published by Digital Divide Data on Hugging Face, will be used for our group project. The dataset contains 56,700 paired speech-text instances, which totals approximately 134.6 hours of audio, with an average clip length of 8.54 seconds (SD = 3.37s). The audio format is .wav and the sources of the audio are from 8 native Cambodian speakers (4 females, 4 males).

- Khmer ASR Cultural Dataset: https://huggingface.co/datasets/Digital-Divide-Data/khm-asr-cultural
- Each row includes:
    - speaker_id: anonymized identifier for one of 8 native speakers

- topic / subtopic: the recording's cultural theme
- transcript: khmer-script text
- audio: the corresponding wav recording
- duration: clip length in seconds

5. **Expected Outcomes**

This project aims to build a functioning Khmer-text-to-speech (TTS) system that is capable of converting written Khmer text into spoken audio. We will use the Khmer ASR Cultural Dataset as we have described above. To obtain constant, consistent speech output, we will train on a single-speaker dataset as opposed to all 8 speakers, which are 4 female and 4 male voices that we can train on. By the end of this project, we will have a well-trained model and a live demonstration where a typed sentence is converted into spoken audio. Since the dataset transcript focuses on Khmer cultural themes, we anticipate the model will perform best within the same domain, and we will discuss this limitation as part of our analysis.

6. **Evaluation Metrics**

In order to evaluate the performance of our Khmer TTS system, we will use a combination of objective and subjective metric standards in TTS research:

- **Mean Opinion Score (MOS)**: Native Khmer speakers will evaluate the generated audio for naturalness and intelligibility using a 1-5 rating scale, following the ITU-T P.800 listening-quality evaluation method (ITU-T, 1996).
- **Character Error Rate (CER)**: Generated audio will be transcribed back into text using an existing Khmer ASR model, and the results are compared with the original text using Character Error Rate (CER). CER is preferred over Word Error Rate (WER) for Khmer given the lack of consistent word boundaries in the script.
- **Baseline comparison**: Results will be [wait for #2 & #3]
- **Low-resource context**: As we have known, Khmer is such a low-resource language with limited paired speech-text data compared to other languages like English. We will interpret our CER and MOS results relative to what is achievable given this data constraint rather than against a high-resource-language TTS system.

7. **References**

Digital Divide Data. (2024). *Khmer ASR Cultural Dataset*. Hugging Face

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